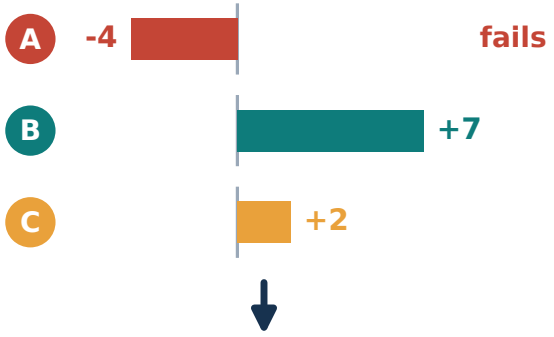
Don’t compress a material system into one depletion date

A typed ledger separates balance, conservation, safety, closure — and the meaning of “time”.

1 Scalar compression

Schematic component balances



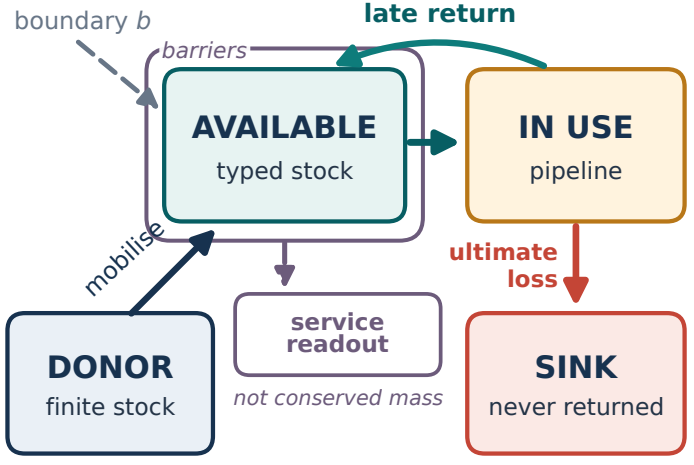
weighted sum $= +5$

one score · one date · one verdict

The aggregate can pass while a critical component fails.

2 Keep the ledger typed

$$\dot{x} = S_{\mathcal{T}} v + b, \qquad v \geq 0$$



substance × place × unit

late ≠ lost

3 Certify separately

BALANCED
the declared balance law holds

CONSERVED
$\ell^{\mathsf{T}} S_{\mathcal{T}} = 0$ for declared moieties

SAFE
every component stays within its barriers

CLOSED
a feasible cycle at declared capacity

No free implications
cross-layer claims need stated hypotheses

“Depletion time” is not one object:

Gross turnover intensity
a pressure scale

Frozen-rate ratio
a local quotient

Scenario hitting time
declared law + barrier